

Feb 10 2026

Ex. $C^0[a,b]$: the set of all continuous functions $[a,b] \rightarrow \mathbb{R}$ with norm:

$$\|x\|_C = \max_{t \in [a,b]} |x(t)|$$

Alternative: $\|x\|_C := \int_a^b |x(t)|^2 dt$

Definition: A normed vector space X is Banach if every Cauchy sequence converges to an element in X

Ex. A converging sequence where the "limit" does not belong to the set

$$a_n = \frac{1}{n}, n=1,2,\dots$$

$$[0,1] \quad a_n \in (0,1] \forall n$$

$$\text{But } \lim_{n \rightarrow \infty} a_n = 0 \notin (0,1]$$

Cauchy Sequence

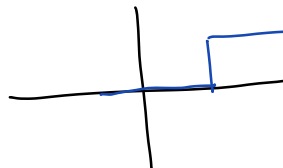
A sequence $\{a_n\}$ is said to be Cauchy if $|a_n - a_m| \rightarrow 0$ as $n, m \rightarrow \infty$

Every convergent sequence is Cauchy.

but not every Cauchy sequence is convergent

A sequence $\{a_n\}$ is called convergent to a^* if for every $\varepsilon > 0$, there exists $N(\varepsilon)$ s.t.

$$|a_n - a^*| < \varepsilon \quad \forall n > N(\varepsilon)$$



Claim: $C^0[a,b]$ is Banach

Back to CT system (existence and uniqueness)

$$\dot{x} = f(t, x) \quad x(0) = x_0$$

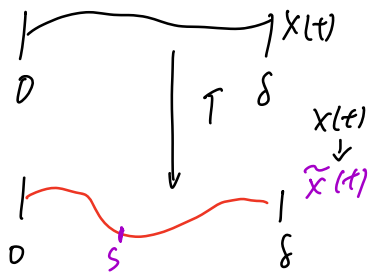
-(*)

we want to show existence and uniqueness of $x(t) \in [0, \delta]$ that satisfies (*)

$$\rightarrow x(t) = \underbrace{x(0) + \int_0^t f(\tau, x(\tau)) d\tau}_{T(x)(t)}$$

DT systems

$$\begin{aligned} x[k+1] &= f[k, x[k]] \\ &= f[k, f[k-1, x[k-1]]] \\ &\vdots \\ &= f[k, f[k-1, \dots f[0, \dots]]] \end{aligned}$$



$$\tilde{x}(s) := x_0 + \int_0^s f(\tau, x(\tau)) d\tau$$

$$\tilde{x}(s) = T(x)(s)$$

$$x(t) = \underbrace{x_0 + \int_0^t f(\tau, x(\tau)) d\tau}_{T(x)(t)}$$

$$T(x)(t) \Leftrightarrow x = T(x)$$

let's investigate application of contraction mapping to $T(x)$

$$(3) \quad U \xrightarrow{T} U$$

$$(1) \quad U(\delta) \subseteq C^0[0, \delta] : \text{Banach space}$$

$$(2) \quad (\text{contraction})$$

$\Downarrow$

$\exists$ a unique $x^* \in U(\delta)$ such that

$$x^* = T(x^*) \quad \text{and} \quad \dots$$

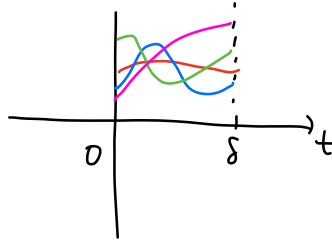
$$U(\delta, r) = \{x \in C^n[0, \delta] : \max_{t \in [0, \delta]} |x(t) - x_0| < r\}$$

closed

$\Downarrow$

(1) is satisfied

$C^n[0, \delta]$



$\bigcup_{C^n[0, \delta]} \bigcap_{C^n[0, \delta]} U$: a sufficient condition is $f(t, x)$ is continuous in t, x

Ex. $f(t, x) = x + u(t)$

$\uparrow$
could be discontinuous on x

Need to show this: $\max_{t \in [0, \delta]} |T(x)(t) - x_0| < r$

$$\underbrace{T(x)(t) - x_0}_{=0} = \int_0^t f(\tau, x(\tau)) d\tau$$

$$x_0 + \int_0^t f(\tau, x(\tau)) d\tau$$

$$\hookrightarrow \int_0^t f(\tau, x(\tau)) d\tau - \int_0^t f(\tau, x_0) d\tau + \int_0^t f(\tau, x_0) d\tau$$

$$|T(x)(t) - x_0| < \int_0^t |f(\tau, x(\tau)) - f(\tau, x_0)| d\tau + \int_0^t |f(\tau, x_0)| d\tau$$

Suppose $f(t, x)$ is Lipschitz in x , uniformly in $t \in [0, \delta]$

$$\Rightarrow |f(t, x) - f(t, y)| \leq L |x - y| \quad x, y \in B_r(x_0)$$

$$< L |x - y| \quad \forall t \in [0, \delta]$$

$f(t, x_0)$ is bounded say by h , because $f(t, x_0)$ is continuous over $[0, \delta]$

$$|T(x)(t) - x_0| < \int_0^t L |x(\tau) - x_0| d\tau + \int_0^t h d\tau$$

$$\begin{aligned}
&< \int_0^t L_r d\tau + \int_0^t h d\tau \\
&= L_r t + h t \\
&< (L_r + h) \delta \\
&< r \quad (\text{want})
\end{aligned}$$

$$\text{Need } \frac{(L_r + h)\delta}{r} < 1 \Rightarrow \delta \leq \frac{r}{L_r + h}$$

$$\rightarrow (2) \quad |T(x) - T(y)| < \rho |x - y|, \quad \rho < 1$$

$$\forall x, y \in U(\delta, r)$$

$$x \in U(\delta, r) \rightarrow |x - x_0| < r$$

$$|x - y| = |x - x_0 + x_0 - y|$$

$$< |x - x_0| + |y - x_0|$$

$$< 2r$$

⋮

to be continued

Ex. $\dot{x} = x^{1/3}$
 $= f(x)$

$$\frac{df}{dx} = \frac{1}{3} x^{-2/3} \Big|_{x=0} = \infty$$

Not Lipschitz

